

Phase Transitions in Random Forests

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Abstract

The arboreal gas is a probability measure on forest subgraphs in which each forest is weighted by $\beta^{|E(F)|}$ where $\beta > 0$. This measure is equivalent to Bernoulli bond percolation with parameter $p(\beta) = \frac{\beta}{1+\beta}$, conditioned on the subgraph being a forest. Our main question is one of phase transition with respect to component structure in the arboreal gas. Does a giant tree exist and, if so, at what value of β does it emerge? We notice that the β -arboreal gas is stochastically dominated by $p(\beta)$ -Bernoulli percolation, which allows us to understand the arboreal gas when our equivalent percolation is subcritical. What is harder to study is the regime in which percolation is supercritical. A giant connected component emerges in percolation, but we are none the wiser as to what is happening in the arboreal gas. In this report we use a technique called the credit network method to estimate phase transitions. It allows us to find the critical point at which the giant tree emerges on the complete graph, aided by results from percolation theory. We discuss extending these methods the more general expander graphs, in particular random d -regular ones, finding an upper bound for the point of phase transition.

1 Introduction

1.1 The Arboreal Gas

We focus on a specific random forest model known as the arboreal gas. Its main phenomenon we are interested in is the existence and emergence of a giant tree. In the context of a finite graph this means a giant tree of positive density exists **whp**. This is an example of a phase transition where a random graph exhibits a dramatic change in typical structure.

Let us elaborate on what some of this terminology means.

Definition 1.1. An event $\mathcal{E}_N$ occurs with high probability (**whp**) if $\lim_{N \rightarrow \infty} \mathbb{P}[\mathcal{E}_N] = 1$.

By positive density we mean that $\frac{|\mathcal{C}(v)|}{N}$ is uniformly bounded away from zero where N is the number of vertices in our graph and $\mathcal{C}(v)$ is the connected component containing vertex v .

We will also frequently use the word order to mean the number of vertices. This could be in the underlying graph or in a connected component. We denote the order of a connected component $|\mathcal{C}(v)|$.

We now define the arboreal gas.

Definition 1.2 (The arboreal gas). Let $G = (\Lambda, E)$ be a finite (undirected) graph and let $\Omega = 2^E = \{A : A \subseteq E\}$. The $\beta > 0$ arboreal gas is the probability measure $\mathbb{P}_\beta$ on 2^E that assigns probability $\mathbb{P}_\beta[A] = \frac{\beta^{|A|} \mathbb{1}_{\{A \in \mathcal{F}(G)\}}}{Z_G(\beta)}$, where $Z_G(\beta) = \sum_{A \in \mathcal{F}(G)} \beta^{|A|}$ and $\mathcal{F}(G) = \{A \subseteq E \mid (\Lambda, A) \text{ contains no cycles}\}$.

We will denote the random spanning subgraph on G due to the arboreal gas as $G_\beta = (\Lambda, A)$.

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1.2 Percolation

Percolation is a model closely linked to random graphs see [Gri99]. We will use knowledge of percolation phase transitions as a tool throughout this report to learn about the giant tree in the arboreal gas.

Definition 1.3 (Bernoulli percolation). *Let $G = (\Lambda, E)$ be a finite (undirected) graph and let $\Omega = 2^E = \{A : A \subseteq E\}$. The Bernoulli $p \in [0, 1]$ percolation probability measure $\mathbb{Q}_p$ on 2^E assigns probability $\mathbb{Q}_p[A] = p^{|A|}(1-p)^{|E \setminus A|}$.*

We denote the random spanning subgraph on G due to percolation as $G_p = (\Lambda, A)$.

Remark. *In the conventional presentation of percolation edges are open and closed. However because we are only using percolation as a tool to study the arboreal gas it seems better to present them in exactly the same way - as probability measures with a sample space of subsets of edges. The random graph they induce is the full set of vertices and the chosen subset of edges.*

The particular phase transition we are interested in for the percolation model is the emergence of a positive density connected component **whp**. This emergence, slightly confusingly, is referred to as *percolation* occurring or we might say that a giant connected component *percolates*.

We notice that in percolation the existence of each edge is independent, whereas in the arboreal gas they are far from it. The singular conditioning on no cycles loses the independence. In fact, it is conjectured that the incidence of two distinct edges is negatively correlated. This collapse of independence means typical methods for proving phase transitions in percolation do not work or are hard to apply to the arboreal gas. For example, the most common method, coupling a breadth first search and a branching process, relies on identifying an offspring distribution. In the arboreal gas the offspring distributions aren't independent. They are correlated, suggesting the branching process method will not work. Therefore, we will have to develop other methods to learn about the arboreal gas.

We make the following useful observation that the arboreal gas is percolation conditioned on there being no cycles.

Proposition 1.1. *Let $G = (\Lambda, E)$ and $p(\beta) = \frac{\beta}{1+\beta}$,*

$$\mathbb{P}_\beta[A] = \mathbb{Q}_p[A | (\Lambda, A) \text{ contains no cycles}] \text{ for all } A \subseteq E.$$

Proof. Rearranging, $\beta = \frac{p}{1-p}$. Thus,

$$\begin{aligned} \mathbb{Q}_p[A | (\Lambda, A) \text{ contains no cycles}] &= \frac{\mathbb{Q}_p[A \cap (\Lambda, A) \text{ contains no cycles}]}{\mathbb{Q}_p[(\Lambda, A) \text{ contains no cycles}]} = \begin{cases} \frac{\mathbb{Q}_p[A]}{\mathbb{Q}_p[\mathcal{F}(G)]} & \text{if } A \in \mathcal{F}(G) \\ 0 & \text{if } A \notin \mathcal{F}(G). \end{cases} \\ \text{For } A \in \mathcal{F}(G), \frac{\mathbb{Q}_p[A]}{\mathbb{Q}_p[\mathcal{F}(G)]} &= \frac{p^{|A|}(1-p)^{|E|-|A|}}{\sum_{A \in \mathcal{F}(G)} p^{|A|}(1-p)^{|E|-|A|}} = \frac{\beta^{|A|}(1-p)^{|E|}}{\sum_{A \in \mathcal{F}(G)} \beta^{|A|}(1-p)^{|E|}} \\ &= \frac{\beta^{|A|}}{\sum_{A \in \mathcal{F}(G)} \beta^{|A|}} = \frac{\beta^{|A|}}{Z_G(\beta)} = \mathbb{P}_\beta[A]. \end{aligned}$$

□

We know there is no giant tree when percolation does not occur due to the following fact.

Proposition 1.2 (Sits below). *Percolation stochastically dominates the arboreal gas*

$$\mathbb{P}_\beta \preceq \mathbb{Q}_p.$$

Proof. Our main tool will be the FKG inequality. We will state it here but a proof can be found in [Gri99, 2.2]. The form we use will be: if $\mathcal{A}$ is an increasing event and $\mathcal{B}$ is a decreasing event then $\mathbb{Q}_p[\mathcal{A}|\mathcal{B}] \leq \mathbb{Q}_p[\mathcal{A}]$. Under the percolation measure, $\mathbb{Q}_p$ cycles are an increasing event via monotone coupling [2.1][Gri99]. Therefore, $\mathcal{B} = \{\text{contains no cycles}\}$ is a decreasing event. By Proposition 1.1 and the FKG inequality, $\mathbb{P}_\beta[\cdot] = \mathbb{Q}_p[\cdot | \text{no cycles}] \leq \mathbb{Q}_p[\cdot]$. □

What does this actually mean? $\mathbb{P}_\beta \preceq \mathbb{Q}_p$ if you can define two random graphs G_1 and G_2 on a shared probability space such that:

- (1) The marginal distribution of G_1 is exactly $\mathbb{P}_\beta$.
- (2) The marginal distribution of G_2 is exactly $\mathbb{Q}_p$.
- (3) $G_1 \subseteq G_2$ almost surely.

The exact construction of this coupling is unknown (an interesting open problem) but we know it exists by Proposition 1.2. Due to this coupling though we can think that probabilistically the arboreal gas always sits below percolation. So if an edge exists in the arboreal gas it also exists in percolation (this is of course also a probabilistic statement). So in order to have a giant tree there must be a giant component in percolation. Thus we are not really interested in the regime with no percolation. The relatively much more difficult regime, on which the rest of this report will be focused, is when percolation does occur. Proposition 1.2 does not give us enough information about what is happening in the arboreal gas to determine whether there is a giant tree or not.

1.3 Order notation

In this report we focus on the limiting behaviour as the number of vertices in the graph, N , tends towards infinity. So there are a couple of definitions important to understand in order to state our results. In the following f and g are some functions of N .

Definition 1.4. $f(N) = \mathcal{O}(g(N))$ if there exists a $C > 0$ such that $|f(N)| \leq Cg(N)$ for all $N \geq N_0$ where $N_0 > 0$ is some threshold.

Definition 1.5. $f(N) = \Omega(g(N))$ if there exists a $c > 0$ such that $f(N) \geq cg(N)$ for all $N \geq N_0$ where $N_0 > 0$ is some threshold.

Definition 1.6. $f(N) = \Theta(g(N))$ if $f(N) = \mathcal{O}(g(N))$ and $f(N) = \Omega(g(N))$.

Definition 1.7. $f(N) = o(g(N))$ if for all $c > 0$, there exists a $N_0 > 0$ such that $|f(N)| \leq c|g(N)|$ for all $N \geq N_0$.

2 The Arboreal Gas on the complete graph

2.1 The Erdős–Rényi random graph

The complete graph, K_N , seems an appropriate setting to begin to understand the emergence of a giant tree, or lack thereof, in the arboreal gas. It enjoys the mean field property; every vertex is identical. The counting arguments we use will rely upon this. It is also a useful setting because percolation on the complete graph is very well understood. Arguably the most studied random graph was first introduced to the world by Erdős and Rényi in their seminal paper [ER59]. We use a slightly different presentation introduced by Gilbert in [Gil59].

Definition 2.1 (Erdős–Rényi random graph). Let $G = K_N$. We perform percolation with parameter $p \in [0, 1]$ on G as defined in Definition 1.3. The resulting graph $G_{N,p}$ is known as the Erdős–Rényi random graph.

We write $\mathcal{C}_j$ for the j -th largest component in $G_{N,p}$.

Erdős and Rényi discovered that this graph exhibits a phase transition at $p = \frac{1}{N}$. A macroscopic connected component of linear order emerges.

Theorem 2.1. Let p be scaled as $p = \frac{c}{N}$ in the Erdős–Rényi random graph $G_{N,p}$. When $c < 1$, $|\mathcal{C}_1| = \Theta(\log N)$ *whp*. We refer to this as the subcritical regime. When $c > 1$, $|\mathcal{C}_1| = \Theta(N)$ *whp*. We refer to this as the supercritical regime.

The terminology used in Theorem 2.1 will be used frequently in this report so it is important to give a bit of context. Galton Watson branching processes are the primary tool to analyse how most random graphs behave. The most common method employed is a breadth first search of a connected component which can then be bounded above and below by branching processes, see [AS08, 11.6]. A subcritical branching process ($\mu < 1$) is guaranteed to go extinct quickly thus in the random graph connected components are kept small. In a supercritical branching process ($\mu > 1$) there is a positive probability that it survives so in a random graph a macroscopic connected component emerges. The further intuition, which is formalised in Lemma 2.5, is that the survival probability of the branching process is akin to the probability that a chosen vertex will be in this macroscopic component.

Due to Proposition 1.2 we are interested in the behaviour of the arboreal gas on the complete graph for supercritical p .

2.2 The Giant Tree

So there is a phase transition at $p = \frac{1}{N}$ for percolation on the complete graph, but what about the arboreal gas? Is the emergence of a giant preserved with the introduction of the singular conditioning or does requiring no cycles completely break it? Intuition would surely lean towards the latter. The conditioning changing the location of the phase transition or erasing its existence all together. Yet it doesn't! We state this astonishing fact formally in the following theorem.

Theorem 2.2. *For $G = K_N$ and $\beta = \frac{\tilde{\beta}}{N}$ with $\tilde{\beta} > 1$,*

$$\lim_{N \rightarrow \infty} \mathbb{E}_\beta \left[\frac{\alpha}{N} \right] > 0,$$

where $\alpha = \max_{v \in G_\beta} |\mathcal{C}(v)|$.

This inequality detects a giant tree in the arboreal gas for $\tilde{\beta} > 1$ and large N . This is very similar to the phase transition in the Erdős–Rényi random graph. The similarity being the giant emerges at $\frac{1}{N}$.

2.3 The Credit Network method

In order to prove Theorem 2.2, we introduce a weaker result and then go about strengthening it through what we know about the Erdős–Rényi random graph.

Lemma 2.3. *For $G = K_N$ and $\beta > 0$,*

$$\mathbb{E}_\beta \left[\frac{\alpha}{N} \right] \geq 1 - \frac{2}{\beta N},$$

where $\alpha = \max_{v \in G_\beta} |\mathcal{C}(v)|$.

This inequality detects the giant tree at $\beta > \frac{2}{N}$. This is of the right order in N so acts as an intermediary step towards our target.

We use in our proof a technique we call the credit network method inspired by [GKRZ14]. We consider the total flow of probability between different states of a Markov chain.

Proof. (Lemma 2.3) Let $G = (\Lambda, E)$ with $|\Lambda| = N$ and $|E| = \binom{N}{2} = n$. We define an irreducible Markov chain with a state space of all the possible subsets of edges that induce a forest in G . This Markov chain enjoys detailed balance, with unique stationary distribution being π_β . Or, as we defined it earlier, $\mathbb{P}_\beta$.

The dynamics of this Markov chain are as follows:

- (1) Select $e \in E$ uniformly at random.
- (2a) If $e \in A_t$ then $A_{t+1} = A_t \setminus \{e\}$ with probability $\frac{1}{1+\beta}$, otherwise define $A_{t+1} = A_t$.
- (2b) If $e \notin A_t$ and $(\Lambda, A_t \cup \{e\})$ a forest then $A_{t+1} = A_t \cup \{e\}$ with probability $\frac{\beta}{1+\beta}$, otherwise define $A_{t+1} = A_t$.

First we must check for detailed balance.

Let $\mathcal{F}(G) = \{A \subseteq E \mid (\Lambda, A) \text{ a forest}\}$ and $\mathcal{F}(G)^{(k)} = \{A \in \mathcal{F}(G) \mid (\Lambda, A) \text{ contains } k \text{ trees}\}$. Fix $A \in \mathcal{F}(G)^{(k+1)}$ and $A' \in \mathcal{F}(G)^{(k)}$ then:

$$\pi_\beta(A)P_{A,A'} = \frac{\beta^{|A|}}{Z_\beta} \frac{1}{N} \frac{\beta}{\beta+1} = \frac{\beta^{|A'|}}{Z_\beta N(\beta+1)} = \frac{\beta^{|A'|}}{Z_\beta} \frac{1}{N} \frac{1}{\beta+1} = \pi_\beta(A')P_{A',A}. \quad (1)$$

So detailed balance holds.

Let $\mathbf{t} = (t_1, t_2, \dots, t_k)$ with $t_1 \geq t_2 \geq \dots \geq t_k$ be the number of vertices of the trees in some forest $(\Lambda, A) \in \mathcal{F}(G)^{(k)}$. Notice that $t_1 = \alpha$.

We want to harness the power of detailed balance by thinking about the flow of probability across the Markov chain. First we think about the adding an edge direction. Let $A \in \mathcal{F}(G)^{(k)}$ with the orders of trees in (Λ, A) being $\mathbf{t}$.

$$\mathbb{P}_\beta[A \cup \{e\} \in \mathcal{F}(G)^{(k+1)}] = \sum_{i=1}^k \frac{t_i(N-t_i)}{2n} = \frac{N^2 - \sum_{i=1}^k t_i^2}{2n}. \quad (2)$$

We then sum this transition over all possible subsets of edges A and A' . We obtain:

$$\sum_{A \in \mathcal{F}(G)^{(k)}} \sum_{A' \in \mathcal{F}(G)^{(k+1)}} \pi_\beta(A)P_{A,A'} = \frac{\beta}{1+\beta} \sum_{A \in \mathcal{F}(G)^{(k)}} \pi_\beta(A) \mathbb{P}_\beta[A \cup \{e\} \in \mathcal{F}(G)^{(k+1)}] \quad (3)$$

$$= \frac{\beta}{(1+\beta)2n} \sum_{A \in \mathcal{F}(G)^{(k)}} \pi_\beta(A) (N^2 - \sum_{i=1}^k t_i^2) \quad \text{by (2)}. \quad (4)$$

We also note an upper bound, which will be useful later.

$$\sum_{i=1}^k t_i^2 = \alpha^2 + \sum_{i=2}^k t_i^2 \leq \alpha^2 + \alpha \sum_{i=2}^k t_i \quad (5)$$

$$= \alpha^2 + \alpha(N - \alpha) \quad (6)$$

$$= \alpha N. \quad (7)$$

Second we think about the removing an edge direction. Let $A \in \mathcal{F}(G)^{(k)}$ with the orders of trees in (Λ, A) being $\mathbf{t}$. Observe that any edge that we remove will increase the number of trees, since we are in a forest. Thus we can count the number of edges that are possible to remove exactly. A tree with t vertices has $t-1$ edges.

$$\mathbb{P}_\beta[A \setminus \{e\} \in \mathcal{F}(G)^{(k-1)}] = \frac{\sum_{i=1}^{k-1} (t_i - 1)}{n} = \frac{N - (k-1)}{n}. \quad (8)$$

We then sum this transition over all possible subsets of edges A and A' . We obtain:

$$\sum_{A \in \mathcal{F}(G)^{(k-1)}} \sum_{A' \in \mathcal{F}(G)^{(k)}} \pi_\beta(A)P_{A,A'} = \frac{1}{1+\beta} \sum_{A \in \mathcal{F}(G)^{(k-1)}} \pi_\beta(A) \mathbb{P}_\beta[A \setminus \{e\} \in \mathcal{F}(G)^{(k)}] \quad (9)$$

$$= \frac{1}{(1+\beta)n} \sum_{A \in \mathcal{F}(G)^{(k-1)}} \pi_\beta(A) (N - (k-1)) \quad \text{by (8)}. \quad (10)$$

We apply detailed balance (1) to (4) and (10) and use our upper bound (7).

$$\sum_{A \in \mathcal{F}(G)^{(k)}} \frac{(N^2 - \alpha N)\beta}{(1+\beta)2n} \pi_\beta(A) \leq \sum_{A \in \mathcal{F}(G)^{(k)}} \frac{(N^2 - \sum_{i=1}^k t_i^2)\beta}{(1+\beta)2n} \pi_\beta(A) \quad (11)$$

$$= \frac{1}{(1+\beta)n} \sum_{A \in \mathcal{F}(G)^{(k-1)}} \pi_\beta(A) (N - (k-1)). \quad (12)$$

We rearrange to neaten things up.

$$\sum_{A \in \mathcal{F}(G)^{(k)}} (N^2 - \alpha N) \pi_\beta(A) \leq \frac{2}{\beta} \sum_{A \in \mathcal{F}(G)^{(k-1)}} \pi_\beta(A) (N - (k-1)). \quad (13)$$

We take an expectation with respect to $\mathbb{P}_\beta$ and use the computational definition of expectation to obtain our desired result.

$$\mathbb{E}_\beta \left[\frac{\alpha}{N} \right] = 1 - \mathbb{E}_\beta \left[1 - \frac{\alpha}{N} \right] \geq 1 - \sum_{k=1}^N \frac{2}{\beta N^2} \sum_{A \in \mathcal{F}(G)^{(k-1)}} \pi_\beta(A) (N - (k-1)) \quad \text{by (13)} \quad (14)$$

$$\geq 1 - \frac{2}{\beta N^2} \mathbb{E}_\beta[N - k] \quad (15)$$

$$\geq 1 - \frac{2}{\beta N^2} N \quad (16)$$

$$= 1 - \frac{2}{\beta N}. \quad (17)$$

□

2.4 The Excess

We introduce a quantity that will be of significant importance in improving our inequality from Lemma 2.3.

Definition 2.2 (Excess). *The excess of a connected component $\mathcal{C}_i$ is defined as $\mu_i = |E(\mathcal{C}_i)| - |\Lambda(\mathcal{C}_i)| + 1$. A graph $G = (\Lambda, E)$ with m connected components has excess $\mu = \sum_{i=1}^m \mu_i = |E| - |\Lambda| + m$.*

Observe that excess 0 and 1 correspond to a tree and a unicyclic component respectively.

In the proof of Lemma 2.3 we made some worst case assumptions that introduced slack into our inequality. This meaning the phase transition is only detected at $\beta = \frac{2}{N}$, which we know to be a factor of two off. Our most notable lazy bounds was $\mathbb{E}_\beta[N - k] \leq N$ (16) where recall k was number of trees in the spanning forest subgraph. This is incredibly pessimistic because even $N - k \leq N - 1$. Our problem is we do not know much about the distribution of the trees in the arboreal gas. Not to worry, we can convert this into things we do know! First, recall Definition 2.2 that $\mu = |E| + |V| - m$. We can rearrange this in the complete graph setting: $N - k = n - \mu$. Our quantity of interest becomes number of edges minus the excess. In the arboreal gas there is **no excess** because it is always a forest. Unfortunately, there is no way to easily compute the expected number of edges. This is because of the unwieldy normalisation constant in the arboreal gas. By Proposition 1.2 though $\mathbb{E}_\beta[n - \mu] \leq \mathbb{E}_p[n - \mu]$ for $p(\beta) = \frac{\beta}{1+\beta}$. So we just need to understand the excess and the expected number of edges in the Erdős–Rényi random graph to find an upper bound for the arboreal gas. Now to make things a bit easier notice

$$p(\beta) = \frac{\frac{\beta}{N}}{1 + \frac{\beta}{N}} \leq \frac{\tilde{\beta}}{N}. \quad (18)$$

By (18) and monotonicity of percolation in p [Gri99, 2.1]

$$\mathbb{E}_{p(\beta)}[n - \mu] \leq \mathbb{E}_{\frac{\tilde{\beta}}{N}}[n - \mu]. \quad (19)$$

To clarify in (19) these are both expectations with respect to $\mathbb{Q}$, the percolation probability measure, just with different parameters.

In Theorem 2.2 $\tilde{\beta} > 1$ and therefore we can exclusively work in the supercritical regime of the Erdős–Rényi random graph. See [AS08, 11.1] for a thorough overview of the regimes.

Informally $G_{N,p}$, when $p = \frac{\lambda}{N}$, $\lambda > 1$, will look as follows:

- $|\mathcal{C}_1| \sim \xi_\lambda N$, where ξ_λ is the survival probability of a Poisson(λ) branching process and the positive real satisfying $e^{-\xi_\lambda \lambda} = 1 - \xi_\lambda$.

- The largest component has excess approaching infinity.
- All other components are trees or are unicyclic.
- $|\mathcal{C}_2| = \Theta(\log N)$.

Our quantity $n - \mu$ is the edges that are not excess edges. Intuition would suggest these comprise of the spanning tree of the giant and almost all the edges in the remainder of the graph. We formalise this intuition in the following lemma.

Lemma 2.4. *Let $p = \frac{\lambda}{N}$ and $\lambda > 1$. Then*

$$\mathbb{E}_p \left[\frac{n - \mu}{N} \right] = \frac{\lambda}{2}(1 - \xi_\lambda)^2 + \xi_\lambda + o(1).$$

To prove Lemma 2.4 we need some intermediary results. The first is the order of the giant.

Lemma 2.5. *When $\lambda > 1$, the largest component is unique and $|\mathcal{C}_1| = (\xi_\lambda + o(1))N$ **whp**.*

Proof. (Lemma 2.5) We summarise the proof in [AS08, 11.9].

Let $\delta > 0$ be a small constant and let $K > 0$ be a large constant. We set $S = K \log N$, $L^- = (\xi_\lambda - \delta)N$ and $L^+ = (\xi_\lambda + \delta)N$. We call a connected component $\mathcal{C}(v)$ small if $|\mathcal{C}(v)| < S$, large if $L^- < |\mathcal{C}(v)| < L^+$ and alien otherwise. From [AS08, 11.9] $\mathbb{Q}_p[\text{there exists an alien component}] = o(N^{-20})$. It follows that **whp** there is no component of order $k \in [S, L^-] \cup [L^+, \infty]$. We call this “no middle ground”.

Set $\gamma = \mathbb{Q}_p[\mathcal{C}(v) \text{ is not small}]$. By the Poisson branching process approximation of the breadth first search [AS08, 11.5] on $G_{N,p}$, $\gamma \sim \xi_\lambda$. Since there is no middle ground, a component not being small is the same as being large **whp**. So any $\mathcal{C}(v)$ is large with probability $\sim \xi_\lambda$ **whp**. Therefore, the expected total number of vertices in large components is $\sim \xi_\lambda N$. However, we still haven’t resolved the issue of there being multiple large components.

We use the idea of sprinkling new edges on $G_{N,p}$ from a graph G_{N,p_1} , where $n^{-2} \ll p_1 \ll n^{-1}$. Our new graph $G^+ = G_{N,p} \cup G_{N,p_1}$ is $G^+ \sim G_{N,p^+}$ where $p^+ = p + p_1 - pp_1$. p_1 is set large enough that at least one of the edges in G_{N,p_1} joins two large components in $G_{N,p}$ with probability $1 - o(1)$. So we have a component of order at least $2\xi_\lambda(1 - \delta)N$ in G^+ . Yet, we have also selected p_1 small enough that $p^+ \sim p = \frac{\lambda}{N}$. By no middle ground, the probability that G^+ has a component this large, an alien, is $o(N^{-20})$. So the probability that G had more than one large component is $o(N^{-20})$. We then make δ arbitrarily small and hence all large components have order $\sim \xi_\lambda N$. We’ve already established that $G_{N,p}$ has expected number $\sim \xi_\lambda N$ of vertices in large components. The contribution to this expectation from G having more than one large component is $o(NN^{-20})$ which is negligible. Hence, **whp**, there is precisely one large component with order $\sim \xi_\lambda N$. $\square$

For the remainder of this section, we refer to the largest component $\mathcal{C}_1$ as the giant and $p = \frac{\lambda}{N}$, unless otherwise stated. The next result is pretty remarkable and is what allows the whole method to come together.

Lemma 2.6 (Discrete Duality principle). [VDH17, Theorem 4.15]

Let $\nu < 1 < \lambda$ be such that $\nu e^{-\nu} = \lambda e^{-\lambda}$. The graph $G_{N, \frac{\lambda}{N}}$ with the giant removed is close in law to the graph $G_{M, \frac{\nu}{M}}$, where $M = \lceil N(1 - \xi_\lambda) \rceil$.

We note that $\frac{\nu}{M} = \frac{\lambda(1 - \xi_\lambda)}{M} \approx \frac{\lambda}{N}$.

Lemma 2.6 is saying that a supercritical Erdős–Rényi random graph with its giant removed looks like a subcritical Erdős–Rényi random graph on the remaining vertices with a tractable probability parameter. This is going to make finding the excess outside the giant in the supercritical regime easier as this is equivalent to finding the excess in a different subcritical graph.

Proof. (Lemma 2.6) See [VDH17, Theorem 4.15] and the references within. $\square$

Lemma 2.7 (Subcritical excess). *When $\lambda < 1$,*

$$\mathbb{E}_p[\mu] = \mathcal{O}(1).$$

This lemma tells us that the excess in the subcritical regime does not grow in N . This is a stronger result than saying that all the components are trees or unicyclic. If there were enough unicyclic components, the excess would grow in N .

Proof. (Lemma 2.7) For branching process arguments in full see [AS08, Chapter 11] or [VDH17, Chapter 4]. The key fact we use is that the progeny of a $\text{Poisson}(\lambda)$ branching process bounds the order of any connected component in $G_{N, \frac{\lambda}{N}}$.

First we make the following useful observation:

$$\sum_{\mathcal{C}} \mu(\mathcal{C}) = \sum_{v \in [N]} \frac{\mu(\mathcal{C}(v))}{|\mathcal{C}(v)|} \quad (20)$$

, where the first sum is indexed by the connected components in $G_{N,p}$. The existence of each edge in the Erdős–Rényi random graph is a $\text{Bernoulli}(p)$ random variable so,

$$\mathbb{E}_p[\mu(\mathcal{C}(v))] \leq \binom{k}{2} p \leq \frac{k^2 \lambda}{2N}, \quad (21)$$

when $|\mathcal{C}(v)| = k$.

The progeny of a Poisson branching process follows a Borel distribution [AS08, Theorem 11.4.2].

Thus:

$$\mathbb{Q}_p[|\mathcal{C}(v_0)| = k] \leq \frac{(\lambda k)^{k-1} e^{-\lambda k}}{k!} \leq C_0 k^{-\frac{3}{2}} e^{-\alpha k}, \quad (22)$$

by Stirling's formula $k! \sim \sqrt{2\pi k} \left(\frac{k}{e}\right)^k$ where $\alpha = \lambda - 1 - \log(\lambda) > 0$ and $C_0 > 0$ is some constant. Hence by (20), (21) and (22):

$$\mathbb{E}_p[\mu] = \sum_{v \in [N]} \mathbb{E}_p \left[\frac{\mu(\mathcal{C}(v))}{|\mathcal{C}(v)|} \right] \leq \sum_{v \in [N]} \sum_{k=1}^N \frac{k^2 \lambda}{2N} \frac{1}{k} C_0 k^{-\frac{3}{2}} e^{-\alpha k} \leq \frac{1}{2} \sum_{k=1}^N C_0 \lambda k^{-\frac{1}{2}} e^{-\alpha k} \leq C(\lambda). \quad (23)$$

Thus the expectation of the excess is bounded above by some function of λ as required. $\square$

Lemma 2.8 (Supercritical excess). *When $\lambda > 1$,*

$$\mathbb{E}_p[\mu(G_{N,p} \setminus \mathcal{C}_1)] = \mathcal{O}(1).$$

Proof. (Lemma 2.8) By Lemma 2.6 $G_{N,p} \setminus \mathcal{C}_1$ is close in law to $G_{M, \frac{\nu}{M}}$ where $M = \lceil N(1 - \xi_\lambda) \rceil$. Since $\nu < 1$, $G_{M, \frac{\nu}{M}}$ is subcritical. We apply Lemma 2.7 which says that $\mathbb{E}_{\frac{\nu}{M}}[\mu] = \mathcal{O}(1)$. $\square$

Now we have the results we need to prove Lemma 2.4.

Proof. (Lemma 2.4) Consider the quantity $n - \mu$ in $\mathcal{C}_1$ and its complement $G_{N,p} \setminus \mathcal{C}_1$, which we will refer to as $\mathcal{C}'_1$. By Lemma 2.5,

$$|E(\mathcal{C}_1)| - \mu(\mathcal{C}_1) = \xi_\lambda N(1 + o(1)) - 1 \text{ whp.} \quad (24)$$

These are just the edges in the spanning tree of the giant. By Lemma 2.6 $\mathcal{C}'_1$ is close in law to $G_{M, \frac{\nu}{M}}$. Hence:

$$\mathbb{E}_p[|E(\mathcal{C}'_1)|] = \binom{M}{2} \frac{\nu}{M} = \frac{\lambda}{2} (1 - \xi_\lambda)^2 N + \mathcal{O}(1). \quad (25)$$

By Lemma 2.8 and (25),

$$\mathbb{E}_p[|E(\mathcal{C}'_1)| - \mu(\mathcal{C}'_1)] = \frac{\lambda}{2} (1 - \xi_\lambda)^2 N + \mathcal{O}(1). \quad (26)$$

We combine the giant (24) and its complement's (26) contributions to obtain:

$$\mathbb{E}_p \left[\frac{n - \mu}{N} \right] = \frac{\lambda}{2} (1 - \xi_\lambda)^2 + \xi_\lambda + \mathcal{O} \left(\frac{1}{N} \right).$$

$\square$

Proof. (Theorem 2.2) First we need to learn about our expression from Lemma 2.4. Recall $\xi_\lambda = 1 - e^{-\lambda\xi_\lambda}$ and hence $\xi'_\lambda = e^{-\lambda\xi_\lambda}(\xi_\lambda + \lambda\xi'_\lambda) = (1 - \xi_\lambda)(\xi_\lambda + \lambda\xi'_\lambda)$. We then have that,

$$\xi'_\lambda [1 - \lambda(1 - \xi_\lambda)] = \xi_\lambda(1 - \xi_\lambda). \quad (27)$$

Let $F(\lambda) = \frac{\lambda}{2}(1 - \xi_\lambda)^2 + \xi_\lambda$. Hence:

$$F'(\lambda) = \frac{1}{2}(1 - \xi_\lambda)^2 - \lambda(1 - \xi_\lambda)\xi'_\lambda + \xi'_\lambda \quad (28)$$

$$= \frac{1}{2}(1 - \xi_\lambda)^2 + \xi_\lambda(1 - \xi_\lambda) \quad \text{by (27)} \quad (29)$$

$$= \frac{1}{2}(1 - \xi_\lambda)(1 + \xi_\lambda) \quad (30)$$

$$= \frac{1 - \xi_\lambda^2}{2}. \quad (31)$$

Since $\xi_\lambda \in (0, 1)$ for all $\lambda > 1$, by (31) $F'(\lambda) > 0$ for all $\lambda > 1$.

Hence, F is strictly increasing in λ for $\lambda > 1^{(*)}$.

We also note that $F(1) = \frac{1}{2}$ (**).

We now improve our proof of Lemma 2.3. We begin from (15).

$$\mathbb{E}_\beta \left[\frac{\alpha}{N} \right] \geq 1 - \frac{2}{\beta N} \mathbb{E}_\beta \left[\frac{N - k}{N} \right] \quad (32)$$

$$= 1 - \frac{2}{\beta N} \mathbb{E}_\beta \left[\frac{n - \mu}{N} \right] \quad (33)$$

$$\geq 1 - \frac{2}{\beta N} \mathbb{E}_p \left[\frac{n - \mu}{N} \right] \quad \text{by Proposition 1.2} \quad (34)$$

$$= 1 - \frac{2}{\beta N} \left[\frac{\lambda}{2}(1 - \xi_\lambda)^2 + \xi_\lambda + o(1) \right] \quad \text{by Lemma 2.4.} \quad (35)$$

We want to know about the behaviour for large N so we take the limit.

$$\lim_{N \rightarrow \infty} \mathbb{E}_\beta \left[\frac{\alpha}{N} \right] \geq \lim_{N \rightarrow \infty} \left[1 - \frac{2}{\beta N} \left[\frac{\lambda}{2}(1 - \xi_\lambda)^2 + \xi_\lambda + o(1) \right] \right] \quad \text{by (35)} \quad (36)$$

$$\geq 1 - \lim_{N \rightarrow \infty} \frac{2}{\frac{\beta}{N} N} \left[\frac{\lambda}{2}(1 - \xi_\lambda)^2 + \xi_\lambda \right] \quad (37)$$

$$\geq 1 - \frac{2}{\tilde{\beta}} F(\lambda) \quad (38)$$

$$\geq 1 - \frac{2}{\tilde{\beta}} F(\tilde{\beta}) \quad \text{by (18)} \quad (39)$$

$$> 0 \text{ for all } \tilde{\beta} > 1 \quad \text{by (*) and (**).} \quad (40)$$

□

This concludes our investigation of the giant tree on the complete graph.

3 The Arboreal Gas on Expander graphs

3.1 Expander graphs

We now discuss a relatively trickier setting for the arboreal gas of expander graphs. Informally we describe expander graphs as finite graphs that are sparse but well connected. They have strong connectivity like the complete graph while having a relatively low vertex degree. This means you can't easily cut off sets of vertices from one another.

One parameter by which we judge if a graph is a good expander is the Cheeger constant. It measures the edge expansion of the graph which is the idea above of it being hard to split the graph in two despite making lots of cuts.

Definition 3.1. (*Cheeger constant*)
A graph $G = (\Lambda, E)$ is a φ_G -edge expander if

$$\inf_{A \subset \Lambda, |A| \leq \frac{|\Lambda|}{2}} \frac{|\partial_E A|}{|A|} =: \varphi_G > 0, \text{ where } \partial_E A := \{(x, y) : x \in A, y \in \Lambda \setminus A\}.$$

3.2 Random d-regular graphs

General expanders are difficult to deal with because their structure can be so variable. So we consider random d-regular graphs G on N vertices, where $d \geq 3$ is a fixed integer and our asymptotics are in N and d . Recall (Theorem 2.2) that on the complete graph the giant tree emerged when the expected average degree in the Erdős–Rényi random graph was about 1 ($p(N-1) \approx 1$). It does not seem completely unreasonable to conjecture that the arboreal gas on other graphs might exhibit a very similar phenomenon.

The result below formalises this conjecture.

Theorem 3.1. For G a random d-regular graph and $\beta = \frac{\tilde{\beta}}{d-1}$ where $\tilde{\beta} > 1$,

$$\liminf_{d \rightarrow \infty} \lim_{N \rightarrow \infty} \mathbb{E}_\beta \left[\frac{\alpha}{N} \right] > 0,$$

where $\alpha = \max_{v \in G_\beta} |\mathbb{C}(v)|$.

This detects the the giant tree for large N and d at $\beta > \frac{1}{d-1}$. We do not know exactly how to prove this result but we offer some ideas for a possible proof.

First we have a weaker result on a general expander:

Lemma 3.2. For $G = (\Lambda, E)$ a φ_G -edge expander with $|\Lambda| = N$,

$$\mathbb{E}_\beta \left[\frac{\alpha}{N} \right] \geq 1 - \frac{2}{\beta \varphi_G},$$

where $\alpha = \max_{v \in G_\beta} |\mathbb{C}(v)|$.

Proof. (Lemma 3.2)(Sketch) We would use the credit network method like in the proof of Lemma 2.3. Instead of exact counting arguments we would use expansion to estimate the edge boundaries of the trees. The Cheeger constant is used as a universal worst case bound. $\square$

This result detects the giant tree for $\beta > \frac{2}{\varphi_G}$. This seems good as there is some reliance on the structure of the graph. It makes intuitive sense that a larger φ_G would make it easier for a giant tree to emerge. More connectivity = more opportunity for components to link. Sparsity = less chance of cycles being formed. Let's examine the strength of the result in a setting we do know about. For large N and d , a random d-regular graph has $\varphi_G = \frac{d}{2}$. So Lemma 3.2 detects the giant tree for $\beta > \frac{4}{d}$. This is a factor of four off, but does have the right order in d . We present two ideas to fix this:

(1) Recall Lemma 2.3 was off by a factor of two and we managed to fix this for large N using the excess on the Erdős–Rényi random graph. For our random d-regular graph we can try something like this, but discrete duality might be harder to find.

(2) We might also seek to specialise past just using φ_G in our argument as this is often a very pessimistic bound for expansion. On the complete graph we did not have to resort to expansion. We could use exact counting arguments. There are more specific facts about the expansion profile on random d-regular graphs which we could make use of [KW14]. The general idea is that on very small sets like the little logarithmic order trees we see the Cheeger constant is a very worst case estimation of their expansion. They are in fact a lot better at expanding than this!

3.3 Next Steps

I am interested in using the full expansion profile of a random d-regular graph to try to improve upon Lemma 3.2 and prove Theorem 3.1. I would use the following ideas:

- Via our stochastic domination result Proposition 1.2 we can think about the orders of the various trees in the arboreal gas as bounded above by connected components in percolation.
- Random d regular graphs are locally treelike **whp**. We can literally evaluate the expansion in a locally treelike region because we can count the edges. We want to say something like all trees in the arboreal gas that are smaller than some $\gamma \log N$ have expansion $\approx d$ in the limit $d \rightarrow \infty$.
- Even for trees that aren't logarithmic in N we can still do better than $\approx \frac{d}{2}$ by utilising [KW14, Corollary 1.4].

I think that by employing these ideas the result of Lemma 3.2 can definitely be improved but maybe not to the extent of actually being able to detect the giant tree at the correct point $\beta > \frac{1}{d-1}$. I think it is important to think of where the losses in information occur in the credit network method and see if we can try and limit these at each step. It is very exciting the idea of finding the point of phase transition on the expander purely through discrete probability just as we have done on the complete graph!

References

- [AS08] Noga Alon and Joel H Spencer. *The probabilistic method*. John Wiley & Sons, 2008.
- [ER59] Paul Erdos and A Renyi. On random graphs i. *Publ. math. debrecen*, 6(290-297):18, 1959.
- [Gil59] Edgar N Gilbert. Random graphs. *The Annals of Mathematical Statistics*, 30(4):1141–1144, 1959.
- [GKRZ14] Ashish Goel, Sanjeev Khanna, Sharath Raghvendra, and Hongyang Zhang. Connectivity in random forests and credit networks. In *Proceedings of the twenty-sixth annual ACM-SIAM symposium on Discrete algorithms*, pages 2037–2048. SIAM, 2014.
- [Gri99] Geoffrey Grimmett. *Percolation*, volume 321. Springer Science & Business Media, 1999.
- [KW14] Brett Kolesnik and Nick Wormald. Lower bounds for the isoperimetric numbers of random regular graphs. *SIAM Journal on Discrete Mathematics*, 28(1):553–575, 2014.
- [VDH17] Remco Van Der Hofstad. *Random graphs and complex networks*, volume 1. Cambridge university press Cambridge, 2017.